Automatic Workspace Calibration Using Homography for Pick and Place

Varun Burde[1], David S. Martinez Lema[2], Vit Zeman[1], Lars Kahlert[1], Tomas Jochman[1] and Pavel Burget[1]

2023 IEEE 19th International Conference on Automation Science and Engineering (CASE) | 979-8-3503-2069-5/23/$31.00 ©2023 IEEE | DOI: 10.1109/CASE56687.2023.10260601

***Abstract*— For deploying any robotic solution, the most cumbersome yet essential task remains calibration. Almost every robotic system uses a camera to extract information to produce some output. In the case of the pick-and-place system, camera information is used to detect objects and estimate their position in real-world space. The accuracy of tasks like sorting and stacking the objects in the box depends on their estimated pose in the space, especially when they need to be arranged in a confined space. We propose the automatic calibration of the workspace using distinct markers. Our experimentation was carried out with loose packets of recycled paper, which consists of small parts inside. We evaluate our results on the basis of translation errors of the estimated centroid position of objects with respect to the position given by the calibrated 3D camera. In conclusion, we also assess the typical pick-and-place performance on the success rate of stacking the packets in a small box.**

I. Introduction

The pick-and-place is a fundamental operation in the automation industry. The human ability to recognize and grasp unknown objects with the dexterous hand motivates automation research. Our research focuses on exploring the Industry 4.0 [1] principles to create a flexible and robust pick-and-place architecture.

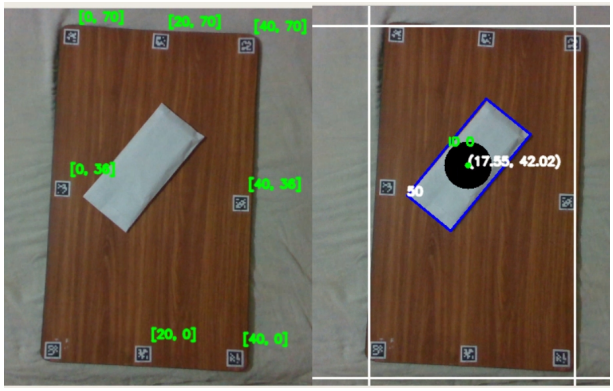
In the era of Industry 4.0, where concepts of Cyber-Physical systems and IoT promise to enhance productivity but they come with new challenges [2]. Recalibration of the workspaces is required to be performed from time to time. Even with minor translation and rotation errors, the workflow can be drastically affected, especially when precision is needed. For example, putting the object in small, confined spaces. Our experimentation is based on such a task where packets must be stacked on top of each other. This scenario is typical in the manufacturing industry, where loose packets are put into boxes for transportation.

When such operating cells are deployed, the camera frame and the robot workspace need to be calibrated to estimate the object's position in the real-world space from the image space. Camera extrinsic calibration [3], [4] is a repetitive and time-consuming task that is inevitable and must be done from time to time.

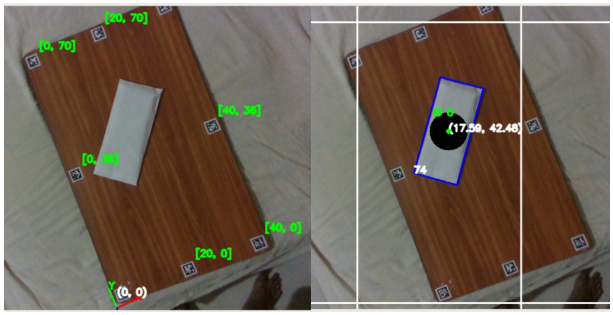
We propose the automatic calibration of the workspace using Apriltags markers [5], which serve as unique identifiers in the image to compute Homographic transformation between the image and the real world. The proposed automatic calibration enables us to achieve pick-and-place operation just from the RGB image without needing the depth.

[1] Czech Institute of Informatics, Robotics and Cybernetics, Czech Technical University in Prague, Prague, Czechia Varun.Burde@cvut.cz ,zemanvit@fel.cvut.cz, kahlelar@fel.cvut.cz , tomas.jochman@cvut.cz , Pavel.Burget@cvut.cz

[2] School of Computation, Information and Technology, Technical University of Munich, Munich, Germany david.martinez@tum.de



(a) The estimated location of packet with homographic transformation



(b) The estimated location of packet with homographic transformation when camera is rotated by few degrees in Z-axis

Fig. 1. Comparison of the position of the packet in marker defined coordinate system when the camera is rotated few degrees in Z-axis (units are in cm).

We propose automatic calibration of the space on conveyor belt, which allows flexible positioning of the camera anywhere in the workspace as can be seen in Fig. 1. Fig. 1(a) and Fig. 1(b) shows different positions of the camera in space but corresponds to the same position in space.

II. Related work

In [6] Chen et al. used homographic transformation to calculate the vehicle's speed to investigate traffic accidents. Their method allows for determining the speed of the vehicle using videos of traffic accidents without any reference to the dimension of the vehicle itself. Using the same method, they

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also extract driving trajectory and velocity curves to better investigate the accident.

To analyze the pedestrian-vehicle activity in a crowded scene, Park et al. presented a framework based on homography [7]. They exploited homography to extract view-invariant object features like footage area and velocity of objects on the ground plane. There presented that their method can be used to enhance situational awareness for disaster prevention, human interaction in a structured environment, and crown movement analysis.

Another application for estimating user's indicator positions using homography for computer vision-based noncontact human-computer systems is shown in [8]. Experiment results show an average error of 0.726mm at a distance of 2.5m when mapping is done using homography. The author also claims that the method's precision and consistency can satisfy the requirement of most applications in human-computer interaction.

In [9] Zhou et al. present an approach to detecting ground employing a monocular sequence captured by an onboard camera using homography. They estimate the dominant homography between two frames taken from the sequence. They claim the proposed method can be used to guide the robot's maneuver as the detected ground plane can be used in obstacle avoidance. The method spans different usage but shows the application of homographic transformation for estimating spaces.

Another example computes the homography between spherical and perspective images [10]. Li et al. showed their experiment results in a visual surveillance system and full-view drone as an application that shows the application of homographic transformation can be used on curved image space.

Yang et al. [11] propose a digital twin-driven measurement method for partial occlusion flexible printed circuit (FPC) assembly. They show robot's involvement in mobile assembly tasks, where they are considered trustworthy because of the small dimension of the parts and the partition occlusion of the robot end effector in operation. They used a homography-based strategy to calibrate the FPC and virtual placement panels in the digital twin.

III. Methodology

Let us assume a conveyor belt is a plane, and point p, which exists in that plane, is given by

$$p =< p_x, p_y >\in \mathbb{R}^2$$

whose corresponding detected point in the image is given by

$$i =< i_x, i_y >\in \mathbb{R}^2.$$

A homographic transformation[12] exists that transforms the image plane into a real-world plane.

$$\lambda \begin{bmatrix} p_x \\ p_y \\ 1 \end{bmatrix} = \mathbf{H} \begin{bmatrix} i_x \\ i_y \\ 1 \end{bmatrix}$$

$$\lambda \begin{bmatrix} p_x \\ p_y \\ 1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} i_x \\ i_y \\ 1 \end{bmatrix}$$

Where λ is scale factor, $\lambda \in \mathbb{R} \setminus 0$.
The matrix in equation form can be rewritten as

$$\lambda p_x = h_{11} i_x + h_{12} i_y + h_{13} \tag{1}$$

$$\lambda p_y = h_{21} i_x + h_{22} i_y + h_{23} \tag{2}$$

$$\lambda = h_{31} i_x + h_{32} i_y + h_{33} \tag{3}$$

To solve for $\mathbf{H}$ consider n image points given by

$$i^n =< i_x^n, i_y^n >\in \mathbb{R}^2.$$

and n points in plane given by

$$p^n =< p_x^n, p_y^n >\in \mathbb{R}^2.$$

Substituting λ from Eq.3 in Eq.1 and 2 and converting to matrix form

$$\begin{bmatrix} i_x^1 & i_y^1 & 1 & 0 & 0 & 0 & -p_x^1 i_x^1 & -p_x^1 i_y^1 & -p_x^1 \\ 0 & 0 & 0 & i_x & i_y & 1 & -p_y^1 i_x^1 & -p_y^1 i_y^1 & -p_y^1 \\ & & & & & \vdots & & & \\ i_x^n & i_y^n & 1 & 0 & 0 & 0 & -p_x^n i_x^n & -p_x^n i_y^n & -p_x^n \\ 0 & 0 & 0 & i_x & i_y & 1 & -p_y^n i_x^n & -p_y^n i_y^n & -p_y^n \end{bmatrix} \begin{bmatrix} h_{11} \\ h_{12} \\ h_{13} \\ h_{21} \\ h_{22} \\ h_{23} \\ h_{31} \\ h_{32} \\ h_{33} \end{bmatrix} = \mathbf{0}$$

This equation can be solved through the Least square method using singular value decomposition, or more robust methods like RANSAC [13] can be utilised.

Because the homographic transformation $\mathbf{H}$ has 8 degrees of freedom, at least four correspondences are needed to compute the solution. Furthermore, the corresponding image points must not be collinear, as this would lead to a degenerate configuration[12]. The homographic transformation is the perspective transformation between two planes, which can compensate for scale, shears, rotation, translation, and perspective distortion. [14] shows the detailed study of Homography and its related error.

The plane points P can be preset on the conveyor belt using distinct markers with the known distance between them. Image points would be the corresponding points detected in the images. The object points detected in the image can be translated to real-world space using matrix $\mathbf{H}$.

IV. Experimentation

A. Hardware Setup

The experimentation setup consists of an Intel RealSense D435 stereo camera above a conveyor belt, rotary encoder for reading conveyor belt speed and position, and a KUKA KR6 robot, all enclosed inside the robotic cell. Fig. 2 shows the perspective view of the setup with the robot beside the conveyor belt and camera.

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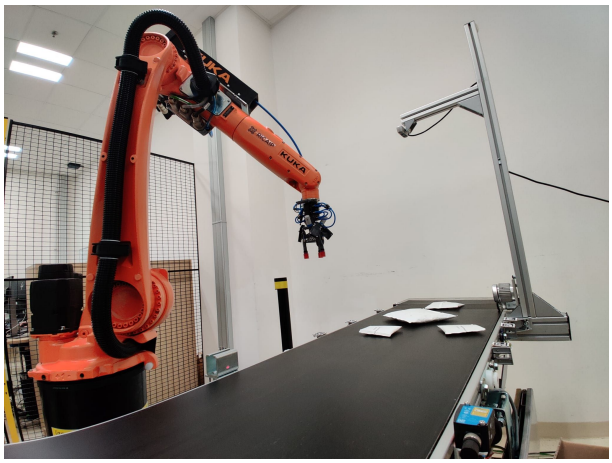

Fig. 2. Physical setup of the robotic cell.

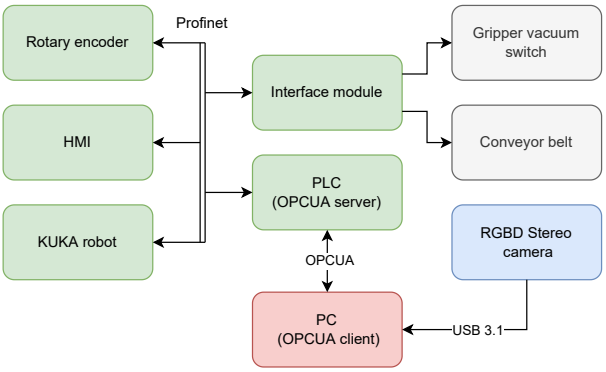

Fig. 3. Interface of hardware to various components.

1) PC: Fig. 3. describes the hardware interface and communication protocols between devices in the cell. The computer is equipped with Nvidia RTX 3060 GPU and Intel 11th gen i7 CPU and performs object detection and tracking. It is connected to the PLC via an Ethernet cable and communicates via OPC UA.

2) Conveyor belt: Our setup consists of an Industry standard conveyor which moves at a constant speed of $0.12\,\mathrm{m\,s^{-1}}$. This is a typical factory routine from the feeder machine to the conveyor belt. Fig. 4. shows the camera's field of view to perceive the moment on the conveyor.

3) Apriltags: We use AprilTags to estimate the plane covering the conveyor pick area by placing them around the edges of the non-moving part of the conveyor system. Rather than extrinsic calibration of the camera, Robot's new base was set up to correspond to the estimated homographic base where one of the tags is considered as the origin. These markers can be placed freely as long as they lie on the same plane. The distances between the markers were measured using a calibrated ruler with respect to the origin.

4) Intel RealSense Camera: The camera used in the project is an Intel RealSense D435. It provides RGB output as well as depth values which are acquired from the stereo camera. It provides depth accuracy of less than two % [15], which is considered as our ground truth in the experiment. We did the extrinsic calibration of the camera with respect to the newly established base created mentioned in IV-A.3 for the ground truth data.

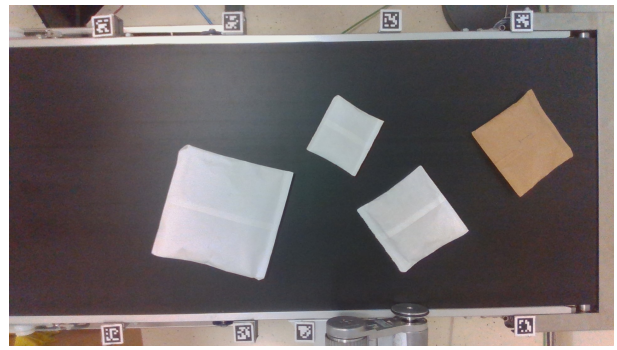

Fig. 4. Camera field of view of the conveyor belt and packets. The conveyor belt is equipped with AprilTags used to calculate the homography transformation.

5) Objects: We demonstrate our experiment with loose square paper packets as mentioned earlier and can be seen in Fig. 4. It consists of four different types of packets: Three white packets of different sizes and one brown packet. The physical dimensions of the packets are 11 $\pm$ 0.5 cm, 14 $\pm$ 1 cm, and 21 $\pm$ 1.5cm on sides and $\pm$ 4cm height.

6) Gripper: The gripper is made up of three suction cups, which are located in an equilateral triangle with the side length of 65 mm. Each suction cup has a diameter of 20 mm. More details about the suction gripper can be found at [16]. The rubber suction cups provide a buffer of 1 cm to compensate for the missing depth dimension.

B. Software Setup

The robot is controlled in External mode and interfaced with the PLC using function blocks provided by the mx-Automation library. This allows process abstraction of the low-level robot control and provides higher-level control via the PLC.

1) Packet Detection: The computer runs Python code which periodically processes images from the RealSense camera. Packets are detected and tracked between images, and once packets leave the camera view, their position is tracked using the information on belt movement provided by the encoder. The program then computes the path of the robot movement for each packet and transfers it to the PLC using the OPC UA protocol.

2) Vision and detection: We used YOLACT [17] to detect and segment the objects. Example of a detected packet and its transformed centroid position with respect to the conveyor belt can be seen in Fig. 5. The detection of AprilTags is made with the OpenCV library [18] and is used to calculate the homography matrix. The homographic transformation between the image space coordinates and real-world coordinates is found with the help of the distances between tags in the image and distances in the real world.

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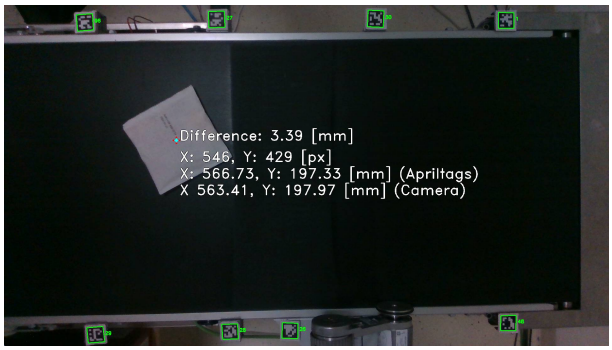


Fig. 5. Image points and their transformed points in real world space.

3) Tracking: An update of the position is necessary to pick up objects correctly. The tracking is divided into two stages according to the object's position on the conveyor belt. The division is based on the camera's field of view. This allows corrections for cases where the belt would stutter, and the position of the object in the camera field of view is corrected. This reduces the number of missed objects.

Packet-tracking is done with the nearest-neighbour algorithm [19], limiting the maximal distance between two objects, after which objects are considered separate entities. Furthermore, if the object is not detected in a defined number of attempts, it is removed from tracking in the camera frame. In that case, the tracking is switched to the second phase, when the knowledge from the rotary encoder updates the position of the object.

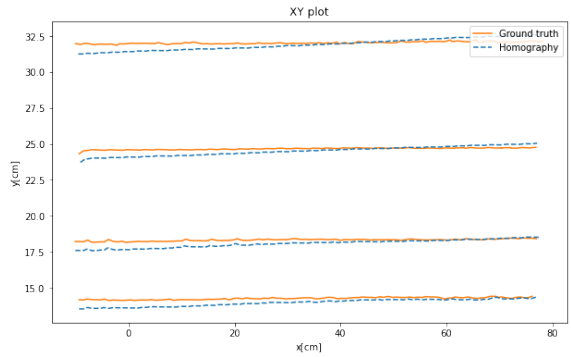
4) Gripping: The robot manipulator tries to match its tool center point (TCP) to the estimated packet centroid for gripping. We use three suction cups to produce the least stress on each branch of the gripper. The height is predefined by the type of the packet.

V. Results

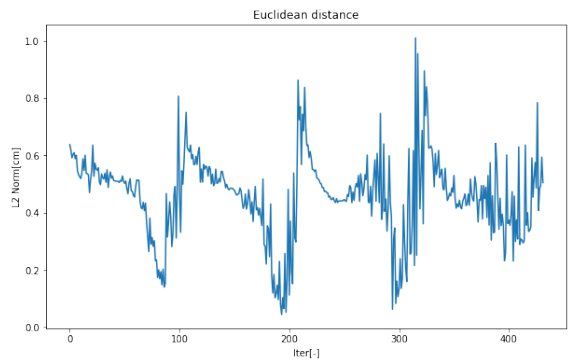
A. Position error in XY plane

The evaluation of two experiments, 1) the movement of packets on the conveyor belt with different starting Y position and 2) the random motion of the packet on the conveyor, are presented in Fig. 6. Fig. 6(a), 6(b) (experiment 1) and 6(c), 6(d) (experiment 2) indicates the estimated position of the packet in the XY axis to the ground truth, and their the euclidean distance error between the estimated centroid point. Fig. 6(b) and 6(d) show that the euclidean distance error between the estimated centroid point and ground truth is always less than 1 cm. Sampled numerical results of experiments can be seen in Table I, II and show that average error never exceeded more than 5mm in each axis.

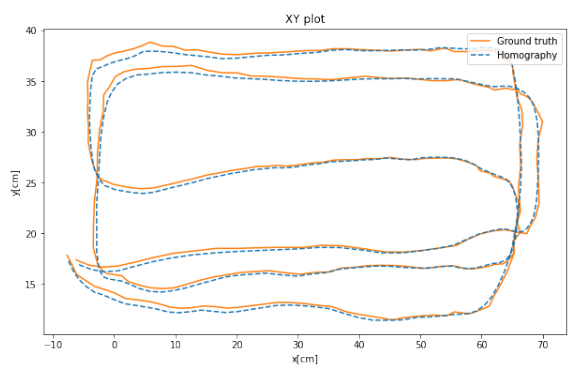
The analysis of the error in positions shows the scaling and misalignment of the plane and can be seen in Fig. 8 (bar graph of the difference between the estimated X-Y location and the ground truth). The X-Y point grid has 10 cm spaces between points, and since the conveyor is longer in the X coordinate, the graph is compressed in that direction.



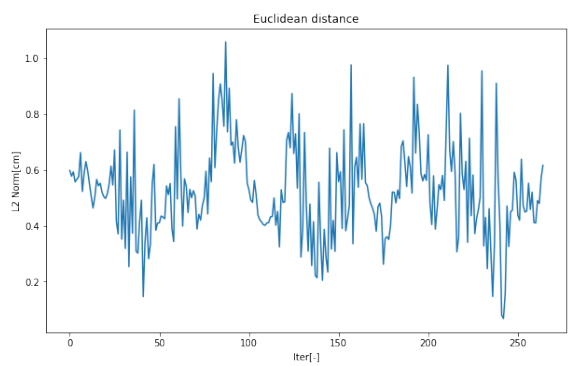
(a) Estimated position of object to ground truth when packets is on top of conveyor belt and follow its motion.



(b) Corresponding L2 norm of the distance between ground truth and the transformed points from Fig. 6(a).



(c) Estimated position of object to ground truth in random motion



(d) Corresponding L2 norm of the distance between ground truth and the transformed points from Fig. 6(c)

Fig. 6. Comparison of estimated XY and ground truth positions in two experiments with error in L2 norm. In Figs. 6(b) and 6(d) the Iter is the sample number taken over shown path to calculate the L2 norm.

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Ground truth centroid [cm]	Homography centroid [cm]	L2 norm [cm]	Absolute distance in x axis [cm]	Absolute distance in y axis [cm]
(-9.3, 13.5)	(-9.3, 14.2)	0.6	0.0	0.6
(28.4, 14.0)	(28.8, 14.3)	0.5	0.4	0.3
(67.0, 14.2)	(66.7, 14.3)	0.3	0.3	0.1
(15.4, 17.9)	(15.7, 18.4)	0.5	0.3	0.4
(53.5, 18.3)	(53.4, 18.4)	0.2	0.1	0.1
(3.7, 24.1)	(3.7, 24.6)	0.5	0.0	0.5
(41.7, 24.6)	(42.1, 24.7)	0.4	0.4	0.1
(-9.5, 31.3)	(-10.0, 32.0)	0.9	0.5	0.7
(28.2, 31.8)	(28.7, 32.0)	0.5	0.4	0.2
(65.1, 32.4)	(65.7, 32.2)	0.6	0.6	0.2
Average distance [cm]		0.5	0.3	0.3

TABLE I

ERROR EVALUATION FOR PACKETS CORRESPONDING TO 6(A).

Ground truth centroid [cm]	Homography centroid [cm]	L2 norm [cm]	Absolute distance in x axis [cm]	Absolute distance in y axis [cm]
(-7.5, 17.3)	(-7.7, 17.8)	0.6	0.2	0.6
(49.5, 11.7)	(49.9, 11.8)	0.5	0.5	0.1
(49.1, 38.1)	(50.0, 38.1)	0.9	0.9	0.1
(-3.8, 28.5)	(-4.3, 29.1)	0.8	0.5	0.6
(42.6, 27.3)	(43.2, 27.3)	0.7	0.7	0.1
(51.4, 16.6)	(51.9, 16.6)	0.5	0.5	0.1
(-2.9, 22.5)	(-3.2, 23.1)	0.6	0.4	0.5
(55.6, 35.2)	(56.4, 35.2)	0.8	0.8	0.1
(43.7, 18.1)	(44.2, 18.2)	0.6	0.6	0.1
Average distance [cm]		0.7	0.5	0.2

TABLE II

ERROR EVALUATION FOR PACKET CORRESPONDING TO 6(C).

Fig. 8(a) shows the difference from the ground truth of values in x-axis, for selected points on the plane, and Fig. 8(b) shows the same difference for values in the y-axis.

B. Stacking of packets

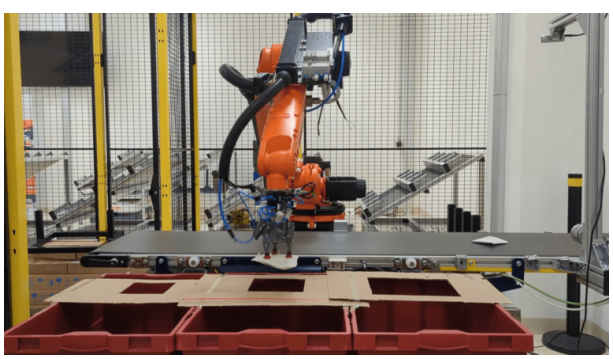
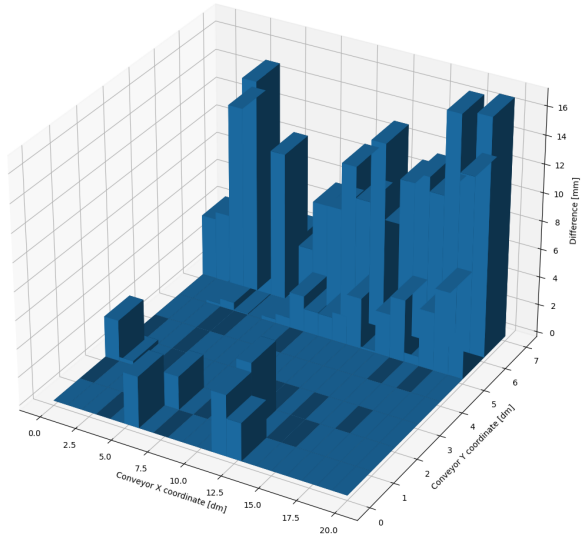
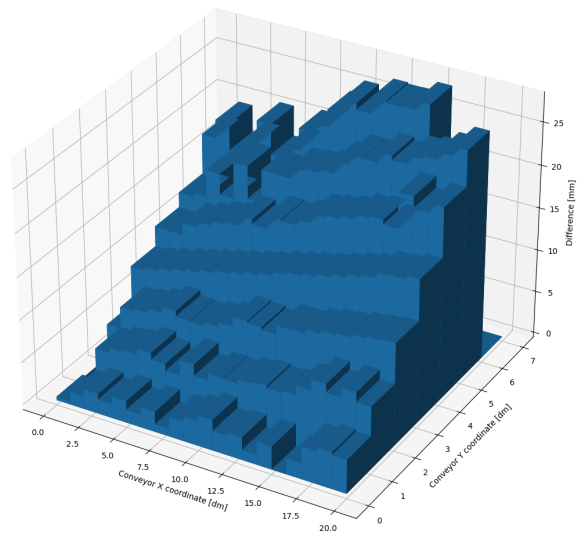


Fig. 7. The experiment setup for the stacking of the packets.

Additionally, we performed the classic pick-and-place and stacking operation in real-time, and the setup can be seen in Fig. 7. We evaluated our results by stacking the packets in the tiny slit. If the packet passes through the hole, it is successful, and if it touches the side of the slit or does not pass through, it is unsuccessful. The slit was made 1 cm longer and wider



(a) X coordinates difference.



(b) Y coordinates difference.

Fig. 8. The difference between X coordinate and Y coordinate and their respective to ground truths, for all points in the X-Y plane.

with respect to the size of the packet mentioned in IV-A.5 on each side to provide flexibility due to the randomness of packet size. The demonstration shows an average of 90.8 % percent success can be seen in Table III.

Furthermore, we calculated the average Takt time of the whole process. Pick and place operation takt time was used to evaluate performance. Takt time was measured as the mean time of 10 consecutive pick and place operations. Each operation begins when the robot starts moving to pick up the packet and ends when the robot returns to some position above the belt and waits for another packet.

VI. CONCLUSION

We successfully demonstrated the usage of markers to gain freedom from the time-consuming calibration process.

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Test number	1	2	3	4	5	Sum
small pass	5	4	7	8	7	31
small fail	3	4	1	0	1	9
medium pass	8	8	7	8	8	39
medium fail	0	0	1	0	0	1
big pass	7	8	8	8	8	39
big fail	1	0	0	0	0	1
Success rate [%]	83.3	83.3	91.7	100	95.8	90.8

TABLE III

SUCCESS RATE OF THE STACKING SMALL, MEDIUM, AND BIG PACKETS IN THE BOX.

In Table III, it can be seen that we achieved a success rate of 90.8 % with a takt time of 3.269 seconds.

In stacking experiment, however every packer passed smoothly from the slit successfully, and only very few hit the edge and were counted as a fail. The cases where the packets were not placed correctly into the slits were mainly caused by tracking errors and errors caused due to position errors shown in Table I and II. The vision algorithm falsely localizes the packet. This happens when the packet is falsely assigned the location of another one in two consequent frames, and the reason is feeder machine draws two or more packets into the conveyor instead of one. Also, the packet can be missed due to the robot's height position not being close enough to the packets. The positional error can happen due to a parallax error during marker placement in the y-axis. Also, the small spikes in Fig. 8 are due to noise in the depth camera and noise in pixels when finding the centroid.

Results demonstrate the flexibility and robustness of the proposed algorithm with minimum error gain and cost reduction of automated cell deployment by removing the depth dependency for simple objects.

VII. ACKNOWLEDGEMENT

This work was supported by Cluster 4.0 – Methodology of System Integration, project number EF16_026/0008432, funded by the Ministry of Education, Youth and Sports of the Czech Republic, Program Operational Program Research, Development, and Education, and by the EU Horizon 2020 project RICAIP (grant agreement No. 857306).

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